

Sensory-friendly Audio and Display Adaptability for Video Playback

Trinity Gao, University of California, San Diego, ngao@ucsd.edu

Jackie Sun, University of California, San Diego, jjsun@ucsd.edu

Michael Schaar, University of California, San Diego, mschaar@ucsd.edu

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1 MOTIVATION

Sensory-friendly audio and display adaptability for video playback are essential areas of research when considering broad potential audiences and the possible needs of users with sensitivities. Nowadays, electronic devices are inseparable from college students. A 2014 study indicated that college students use their cell phones for 8 to 10 hours daily [7], a figure that rose by 20–30 percent during the COVID-19 pandemic, as reported by Meyer in 2020 [4]. Given this prolonged exposure, there is a pressing need for sensory-friendly interfaces, particularly for students with impairments. For instance, individuals with epilepsy may experience seizures triggered by rapid light changes in electronic displays. The potential benefits of such adaptive designs for college students include ergonomic improvement and reduced discomfort from extreme stimuli. However, the practicality and effectiveness of sensory-friendly audio and visual modifications during video playback still need to be explored. Questions also arise on whether to focus on textual warnings or to manipulate the output directly, two intuitive features for sensory-friendly adaptation. To address this, we plan to conduct a literature review and interviews with college students. This report aims to assess the effectiveness, ergonomics, and acceptability of various approaches for sensory-friendly audio and display implementations during video playback. A sensory-friendly design will be more user-friendly for diverse individuals by accommodating their sensory needs when they consume video media.

1.1 Scope

The scope of this report is limited to light and display outputs for video playback. While many user interfaces can negatively affect users sensitive to certain sights or sounds, video playback is a vital area of study concerning this topic. Users often receive little to no warning before the extreme stimuli are presented suddenly, and warnings that do exist are subject to the biases of the video creators. Video playback is also a technology that has become commonplace, and narrowing our focus to this topic will allow us to make valuable observations for future research.

We are also limiting the scope of our research subjects to UCSD college students. This subgroup, mainly composed of young adults who have interacted with video media their entire lives, can provide meaningful feedback for our project and give observations about the nature of extreme stimuli in video playback. Furthermore, this subgroup is also easily accessible for interviews. By targeting this specific demographic, we aim to gather nuanced insights and perceptions from a defined group representing a

diverse yet concentrated segment of the broader college student population. While acknowledging the need to discuss experiences with sensory-friendly design across various user groups, this report intentionally limits its investigation to streamline the research process.

1.2 Background

Technology integration into daily life has bolstered the need for sensory-friendly design. Within the realm of display and audio for video playback, the widespread use of streaming services and video applications on electronic devices, which are generally held closer to the face than televisions and may also require headphones, has also influenced the need for this area of study [7]. As the usage of electronic devices surges, there is an increasing need to recognize the potential impact on individuals who have sensory impairments or are adversely affected [4].

Some strides have already been made to protect users from the extreme stimuli many smart devices produce [2]. Photosensitivity warnings in videos with flashing, strobing lights attempt to accommodate users with conditions such as epilepsy, who may have seizures triggered by viewing such media. Many devices also have features to easily adjust the brightness and volume of videos to fit the user's preferences. Some adjust display brightness automatically based on the user's outside environment [1]. As the technology landscape continues to evolve, so does the understanding of sensory-friendly design. This report aims to evaluate the effectiveness, ergonomics, and acceptability of various implementation approaches for sensory adaptability in video playback through target user interviews and a literature review of related works.

2 RELATED WORK

2.1 Topic 1 - Sensory-friendly Adaptability in Existing Devices

"Sensory-friendly adaptability" is a crucial topic because it allows the advancement of the user experience and caters to the needs of sensitive users. This topic elaborates on how designers have attempted to include sensory-friendly audio and display adaptability in the past. While our other topic focuses more on the overarching societal complexities of electronic devices, this topic indicates how related works have discussed solutions that can enhance ergonomics and effectiveness by warning against extreme stimuli. By focusing on this aspect, we aim to empathize with the user's experience by considering the strengths and weaknesses of previous solutions.

2.1.1 Topic 1, Example 1 - Supporting Noise Sensitivity and Emotion Regulation with Children [3].

Dotch et al. conducted collaborative design sessions with noise-sensitive children to investigate how to design an assistive application to support emotion regulation in children. One finding is that a system supporting noise sensitivity in children should include personalization, gamification, and support for

self- and co-regulation. This allows users to develop new coping mechanisms for noises. Also, personalization features in the system may help with noise sensitivity and emotional regulation by increasing children's engagement.

This work relates to our idea by investigating the design choice in assistive interfaces to support noise-sensitive children. As a contribution to the HCI community, we could use their findings as a reference when assessing different implementations of sensory-friendly adaptation in video playback. Our idea also differs from Dotch's work because we focus on average college students as our primary audience and their potential needs for sensory-friendly adaptation without conducting a design study. We also assess the various approaches for implementing sensory-friendly audio and display during video playback. This could reveal ways to improve user experience during video playback [3].

2.1.2 Topic 1, Example 2 - Designing new interfaces for digital interactive television usable by older adults [7].

Rice et al. discuss the design of user interfaces targeted toward older adults and confront accessibility and usability issues with modern interactive television technology. It focuses on design aspects that can be improved, such as navigational complexity, closed captioning, and graphical output.

This work relates to our idea by discussing the design choices necessary to make audio and light output accessible to extreme users. However, this work also differs in focusing primarily on television, which has limited interactivity options compared to a smart device. Our idea builds upon this work in a way that contributes to HCI by discussing how sensory-friendly interfaces can be applied to modern electronics, broadening the scope of design consideration to include young adults who benefit from sensory-sensitive user design [7].

2.1.3 Topic 1, Example 3 - Adaptive lighting for energy-efficient comfort and well-being [6]

Papamichael examines the feasibility of automatically adjusting the lighting in a room based on its current occupancy and daylight levels. The paper aims to increase energy efficiency and cost-effectiveness in indoor and outdoor spaces. However, the writers also considered the importance of user comfort and well-being and factors such as occupant behavior, privacy, and glare reduction.

The work relates to our idea because it discusses the motivation and design implications of an adaptive lighting system for comfort and well-being. However, this work differs in focusing on architectural mechanisms such as indoor and outdoor lighting instead of electronics. Also, it does not touch upon the same digital well-being concepts that we do. The ideas presented in our report build upon this work by focusing more on electronic programs through the lens of sensory-friendly HCI-inspired user interfaces, building upon this prior work by applying similar principles to a modern piece of technology [6].

2.1.4 Topic 1, Example 4 - iPhone features: Adjustable brightness and color temperature [1].

Adjusting brightness and color temperature on iPhones includes manual brightness adjustment and settings for Dark Mode, True Tone, and Night Shift. The features above enhance the visual comfort of users. The ambient light sensor helps to auto-adjust brightness. True Tone would dynamically match display color and intensity with ambient light, while Night Shift shifts colors to warmer tones for comfortable usage at night. These features aim to reduce eye strain and improve user experience through adaptable display settings.

Our proposed idea focuses on making the display and audio during video playback sensory-friendly for college students instead of displaying modes that work in all contexts. This offers a new HCI perspective to improve user experience by discovering the possible implementations for sensory-friendly audio and display. The iPhone feature gives a current industry example of a sensory adaptable display. We aim to build upon their features and find potential additional features, such as extreme audio filtering, for a better user experience during video playback. [1]

2.1.5 Topic 1, Example 5 - Adaptive Sound and Light Control in Clinical Environments [9]

The SoundLight project by Saiu and Grosso is an adaptive system based on an intelligent agent designed to control and monitor auditory and visual stimuli in clinical settings. The interface automatically processes sound signals and converts them into controlled lighting effects perceived as relaxing by the patients. This alleviates the patient's psychophysiological condition before and after the surgery.

The paper investigates the properties of sound and light waves that induce relaxation in patients, which we could use in our literature review as potential properties for sensory-friendly adaptation. Their focus on controlling and monitoring auditory and visual stimuli is similar to our project. Our idea differs from the SoundLight project by focusing on average college students instead of patients. Also, our project contributes to the HCI community by analyzing potential features that reduce the discomfort of extreme light and sound stimuli for video playback. This lens builds upon the work done in the SoundLight project, which focuses explicitly on display [9].

2.2 Topic 2 - Digital Wellbeing

This topic represents the broader impact and implications of incorporating sensory-friendly adaptability in digital interfaces. While our previous topic considers the technological aspect of the project, this topic acknowledges the societal implications of technology usage through a social perspective. The examples associated with digital well-being focus on the long-term societal effects of technology usage, emphasizing the adverse sensory effects on users. By considering digital well-being in the context of societal impact, we seek to broaden the scope of our project beyond technological solutions, elaborate upon our motivations for benefiting college students, and contribute to a supportive digital ecosystem within society.

2.2.1 Topic 2, Example 1 - Association for Computing Machinery(ACM) Designing Interactive Systems [2]

This conference was designed to provide a place where designers, engineers, psychologists, and many more can come together and join perspectives from various backgrounds on interactive systems design. The fundamental concept of this conference was the idea of “digital well-being,” its goal was to ask and understand how designers can prioritize the well-being of themselves and users when designing a product. They found that most attendees were interested in the idea and submitted 600 papers and pictorials. 131 out of 578 articles and pictorials were accepted, giving them a 23% acceptance rate.

Our idea differs from what was discussed at the conference since we aim to empathize with real-world users through our interviews in addition to focusing on a technological angle. We want to understand how to accommodate users with particular sensitivities and disabilities. We can build off the idea of “digital well-being” by speaking with fellow users and understanding how this perspective can influence our thought process and considerations regarding our design. While this conference's emphasis on digital well-being supplies us with the framework for our area of research, we believe we can build on its scope through real-world empathizing [2].

2.2.2 Topic 2, Example 2 - Changes in physical activity and sedentary behavior due to the COVID-19 outbreak [5].

This article explores the increasing reliance on technology, particularly cell phones, among Americans, especially college students. Through a survey methodology that collects the experiences of college students, the paper highlights the integral role of technology in daily life and how it shapes social interactions. Key findings emphasize the integration of cell phones into students' lives and the potential unawareness of their dependency on these devices.

Our proposed idea diverges from the article by offering insights into how sensory adaptations can positively impact the user experience, potentially influencing the design and implementation of video playback interfaces within educational settings. Our project would offer a practical perspective on creating more inclusive interfaces, substantially affecting the overall digital well-being of college students since many involve electronic devices in their lifestyles. It moves beyond discussing the general trend of technology dependence among college students to actively address a specific aspect that can enhance the user experience and inclusivity within the HCI domain, offering tangible ways to improve digital interactions for a particular group of users [5].

2.2.3 Topic 2, Example 3 - The invisible addiction: Cell phone activities and addiction among male and female college students [8].

This related work aimed to assess the impact of COVID-19-related public health guidelines on various aspects, including physical activity and mental health. It surveyed around 3000 adults from multiple US states through self-reported levels of health markers such as stress levels, loneliness, and positive mental health. Not meeting PA guidelines and increased screen time of adults correlated with decreasing mental health indicators. The study suggests that maintaining or enhancing physical activity and limiting increased screen time during abrupt societal changes might alleviate mental health consequences.

Unlike the mentioned study focusing on the pandemic's impact on behaviors and mental health, our idea is novel in its concentration on improving user experiences through tailored adaptations for individuals with sensory sensitivities. While the overall goal of limiting screen time is outside the scope of our topic, we have realized that digital well-being is a significant concern that involves more caveats than simply focusing on audio or display sensitivities. Our idea contributes to HCI by offering practical solutions for inclusivity within a specific domain, potentially influencing the design and implementation of interfaces in educational settings. This proposed approach addresses a particular need — enhancing accessibility and comfort during video playback — which brings attention to the purpose of digital well-being expressed in this work [8].

2.2.4 Topic 2, Example 4 - Sound-level monitoring earphones with smartphone feedback as an intervention to promote healthy listening behaviors in young adults [4].

Knoetze discusses the prevalence of young adults potentially damaging their hearing by listening to damaging audio outputs from electronic devices. It focuses on hearing well-being and advocates for strong interventions to ensure healthy listening behaviors.

This work relates to our idea because it shows that digital well-being is intrinsically tied to sensory-friendliness. Our idea is novel because the technology described in the paper only warns the users from damaging sound output. We seek to discuss possible contributions to digital well-being through a more holistic lens that scrutinizes adverse display outputs. It's important to consider healthy electronic use from multiple angles, and we have structured our research to analyze various forms of sensory-based adverse effects as a result. This approach contributes new information to the HCI community by considering the interconnected nature of audio and visual extreme stimuli [4].

2.2.5 Topic 2, Example 5 - Self-Conscious or Self-Confident? A Diary Study Conceptualizing the Social Accessibility of Assistive Technology [10]

Shinohara and Wobbrock explore the topic of Assistive Technologies (ATs) and how they can affect disabled users' self-perceptions and social interactions when used in public spaces. They emphasize the idea of “social accessibility” and how important it is to balance functionality and identity. Their methodology revolves around analyzing diary studies of participants with and without disabilities. The users with disabilities will note how they feel when they are using ATs in social contexts, while the users

without disabilities express how they feel when observing a disabled person using ATs in public. Key findings suggest that the AT design substantially impacts self-confidence, self-efficacy, insecurity, and social interactions. As a result, the study offers an angle of entry in Design for Social Accessibility (DSA), which promotes ATs to be designed in such a way that is not only effective at accomplishing the task they were made for but also ensures that social equity and inclusion is a top priority to provide the user with a sense of belonging and confidence.

Our proposed topic is different since we aim to concentrate on digital interfaces and their impact on individuals with sensory sensitivities instead of Assistive Technologies' effect on disabled users in social settings. Rather than focusing on individual experience in public spaces, we emphasize the broader implications of technology use in society. Instead of targeting users with disabilities, we direct our focus on UCSD college students as a whole, even if they don't have any sensory sensitivities. We hope to provide a new perspective on designing to accommodate sensory sensitivities to the HCI community by taking advantage of the principles of social accessibility introduced by Shinohara and Wobbrock, employing them in digital interfaces, and highlighting the importance of sensory-friendly designs in technology [10].

3 METHODS

Our methodology explores our thesis of advocating for the importance of sensory-friendly adaptability in audio and display thoroughly, as we can gather observations from multiple contexts to determine the relevance of our topic and the efficacy of its related solutions. Our research consists of two distinct steps: a literature review of the aforementioned related works, intended to provide a broader understanding of existing knowledge of sensory-friendly adaptability, and an interview portion, directed towards UCSD college students and meant to receive valuable perspectives on people's experiences with video media and its outputs. This dual approach is intended to provide a more accurate representation of our topics, as we can receive valuable information from both well-established researchers and members of our real-world users.

First, our literature review was conducted by analyzing related works with a solid topical focus on sensory-friendly adaptability, as well as those that generally discussed the sensory effects of video media. This step in our methodology became the focus of our research based on feedback from our instructors, who recommended an analysis of pre-existing work to supplement the interview portion. The literature review was an appropriate source of knowledge that would build upon our interviews. We chose to analyze selected papers from related works in the previous section for the literature review. While the related works section provides general background knowledge about our associated topics and only gives a broad overview of the relevant paper without relating it to other articles, a literature review discusses how these works interconnect and forms a more significant thesis based on all of the collected information. We selected six related works that provided solid opportunities for this interdependent analysis. The scope of our study is mainly thematic; we looked for common themes among all of the

literature, whether any of them agreed with or contradicted each other on specific points, and the approaches they used for their methodology.

Next, each of our group members interviewed two UCSD college students, seeking their perspectives on sensory-friendliness through a carefully curated collection of relevant questions. The participant's basic information, including name, gender, and major, is collected to validate participant pool diversity.

Based on feedback from our instructors, we sought to provide unbiased questions, avoiding words with positive or negative connotations. Another point of feedback is that providing extreme stimuli, such as videos with bright lights and loud noises, was outside the scope of our research, as such a study would require ethics training and extremely cautious preparations to avoid adverse effects on the participants. Instead, we opted to provide precise descriptions for the participants to contemplate and answer, each describing a fictional scenario in which the participant views video media with extreme stimuli, and each having differing variations on potency, spontaneity, or both. The first scenario describes a scene where an unexpected, loud, and sharp noise is present when watching lecture podcasts. The second scenario depicts a scene where an unexpected, bright, and white screen is flashed when watching Netflix shows. We ask how they would cope with extreme sensory stimuli without sensory adaptation. Then, we present these two scenarios again with an addition of either a warning or direct output manipulation. The participants then gave pleasantness and effectiveness ratings on each of these scenarios and the reasoning for their rating. We also asked about the participants' opinions on the social acceptability of both sensory adaptation features. Our methodology believes that the perspectives offered by the participants, even through the lens of invented scenarios, would still provide valuable insights into the different ways sensory-friendliness can be achieved and the perceived efficacy of these solutions. Recruited participants were all UCSD college students who had already formed a strong rapport with us, and our questions were provided in a comfortable environment; choosing these kinds of participants was beneficial in ensuring we would receive feedback in a timely and risk-free manner.

Our dual methodology provides reasonable coverage of our topic, as we can gather insight from not only the academic world, which specializes in providing a broad overview of pre-existing knowledge and professional attitudes towards sensory-friendliness, but also from fellow UCSD college students, who can provide more adventitious perspectives that elaborate upon how young people, a relevant user group for our research, utilize and perceive video media. The methodology is feasible for our research group but can still capture enough information to build our thesis, provide valuable results, and generate appropriate discussion.

4 RESULTS

4.1 Literature Review: Key Findings

After a comprehensive review, we can further understand how sensory-friendly design can contribute to the user's overall well-being. We can see a clear connection between individualized sensory adaptability and digital health. One desire frequently encountered across various studies was the need for personalization. Dotch et al.'s work with noise-sensitive children highlights the importance of customizable settings and features to accommodate unique individual sensory needs. Dotch et al. state, "A system supporting noise sensitivity in children should include personalization, gamification, and support for self- and co-regulation." We can understand how important it is to look past general accessibility and adopt a more user-centered approach. As time passes, we notice that a wider variety of user groups are accounted for to ensure a sense of belonging and that no user is left out. It's important to recognize this expansion since diversity in sensory sensitivity and equity amongst users should be one of the top priorities of designers in a user-centric model.

The Association for Computing Machinery (ACM) Designing Interactive Systems conference [2] and the SoundLight System project [9] emphasize the vital connection between sensory adaptability and digital well-being. According to the ACM's Designing Interactive Systems conference [2], "most attendees were interested in digital wellbeing and submitted 600 papers and pictorials. 131 out of 578 papers and pictorials were accepted, giving them a 23% acceptance rate." As technology becomes more prevalent, it's crucial to provide the user with the best digital experience possible and avoid any form of discomfort we can. Regarding ergonomic implications, Shinohara and Wobbrock and Rice et al. findings suggest sensory-friendly designs can significantly impact and benefit users' overall productivity. This is due to a multidimensional effect on the user that encapsulates physical comfort, self-confidence, social accessibility, and equity. One substantial finding across many studies was the desire for individuality and the wide variety of different sensory needs amongst users. Looking forward, we should consider designing adaptable interfaces that cater to users all across the board and account for unique needs and preferences. The idea of digital well-being is championed in studies like Rice et al., Knoetze et al., and Papamichael et al., which all embrace and offer new avenues for a broader application of these more accommodative designs. We must think past accessibility and focus on the importance of creating an equitable and user-friendly digital interface. Now more than ever, the need for diversity and inclusivity in technology is at the forefront of designers and other disciplines worldwide.

4.2 Interview: Statistics, Quotes, and Key Findings

We recruited 6 participants for our interview regarding sensory-friendly adaptation. All participants are from different majors, ensuring diversity of experience. Two participants are not sensitive to strong light or sound at all. Two reported varying degrees of sensitivity depending on stress level or environment. One participant claimed he is sensitive only to light. One participant stated that she is sensory sensitive to light and sound, particularly light with high saturation and brightness and sound with high frequency

and amplitude. Despite our small participant pool, 4 out of 6 participants reported having certain extents of sensitivity to intense light and sound, supporting the motivation of our project.

Some participants rated effectiveness and pleasantness in range instead of a single number, so we decided to use the median of the range for statistical analysis. The average pleasantness without sensory adaptation description for the first scenario is 4.67. 4 participants conveyed the scenario as “wouldn't be bothered too much.” In particular, the participants with sensory sensitivity stated that the scene “makes me feel angry and irritated and triggers my automatic response.” As expected, we observed that intense sound most affects sensory-sensitive people. When asked about the coping strategies for situations where there is unexpected noise, all participants gave vastly different responses, from lowering the volume to messaging the professor. This shows that people's coping mechanisms for extreme sound stimuli are personal.

When the first scenario is illustrated with a warning feature before the onset of extreme stimuli, the average pleasantness responses increased from 4.67 to 6.71. The average effectiveness of this sensory adaptation is 7. Most participants say that the warning effectively removes the surprise factor of the extreme sound, so the user's experience improves. 2 participants mentioned that the warning's timing and design are essential factors since they influence the success rate of mitigating the noise before it occurs. The participants agreed that this feature is socially acceptable depending on the warning's effectiveness, convenience, user sensitivity, and frequency. These design aspects should be considered when implementing the sensory-friendly adaptation for UCSD college students.

Regarding the strategy of directly manipulating the sound output, the pleasantness rating is 7.17, and the effectiveness rating is 7.67. This slight increase in both ratings shows that our participants prefer direct manipulation of sound output over textual warning. 4 participants approve it as “effective in minimizing the disturbance caused by a loud and sharp noise.” In comparison, 2 other participants have doubts about the video experience being affected by direct manipulation. This signifies that user control and freedom are crucial when designing sensory-friendly interfaces. The users should always have the choice to not use this feature during video playback. Overall, participants decided that direct manipulation is socially acceptable if the viewing experience is unaffected and user control is preserved.

For the second scenario, the average pleasantness rating is 4.75. Since the rating is similar to the first scenario, we know that the description for extreme light stimuli is consistent and comparable with that for the sound stimuli. Most participants said the bright screen is “not a pleasant experience.” Unlike the previous scenario, the coping mechanism reached a relative consensus where participants agreed to turn down the brightness and close the electronics as their strategy.

When the second scenario is represented with a warning before the onset of extreme stimuli, the average pleasantness and effectiveness ratings are 5 and 5.5. For their ratings, participants noted that the warning may interfere with their viewing experience while still enduring the flashing screen. Interestingly,

participants all approved that this feature is socially acceptable because “a warning may be helpful for people”; however, 4 participants stated that this feature would be bothersome for themselves after saying that it is socially acceptable. So, the social acceptability of the textual warning for display stimuli requires further investigation.

For direct manipulation of the display output, the pleasantness rating is 7.5, and the effectiveness rating is 7.33, a significant increase compared to the pleasantness and effectiveness rating of textual warning and the baseline scenario without sensory adaptation. 4 participants declared that automatic manipulation of the display output prevents them from the discomfort of a bright screen and leads to a better user experience without manual adjustments. However, 2 other participants showed concern regarding the possible effect on the viewing experience due to visual detail loss from manipulation. All participants agreed that the solution is socially acceptable with suitable implementation. Two participants even expressed that the social acceptability of direct manipulation is higher than that of textual warnings.

Overall, the participants prefer output direct manipulation to textual warning, especially for extreme display stimuli. An improvement in pleasantness can be observed from adding sensory-friendly adaptation to all scenarios, showing that sensory-friendly adaptation fits within the ergonomics of users. Unfortunately, we cannot verify the significance of our statistics due to our minimal sample size. The original coping mechanism is diverse for extreme sound stimuli but not for display stimuli. The sensory-friendly design would benefit UCSD college students in varying degrees, specifically enhancing the user experience of sensory-sensitive students. The design aspects that should be considered when designing a sensory-friendly adaptation interface are effectiveness, convenience, user sensitivity, warning/manipulation frequency, user experience, and user control and freedom. The interface should not interfere with the user's control and experience while ensuring the effectiveness and convenience of adapting adverse sensory stimuli. This would increase the social acceptability and ergonomics of the sensory-friendly adaptation feature for video playback.

5 DISCUSSION

5.1 Limitations and Future Directions

The limitations of our literature review and interview regarding sensory-friendly adaptation include the potential bias of our data and the un-generalizability of our result. Our participant pool is limited to UCSD college students. Our data cannot be generalized to draw solid conclusions about college students. Also, due to the time and resource constraints, our team can only recruit 6 participants and analyze 10 sources, which is a significantly low sample size. An insufficient sample size would mean that our participants and literature may be non-representative of our population and topics and increase our findings' potential biases. However, our interview findings proposed meaningful directions for future research in sensory-friendly adaptation for UCSD college students. Therefore, further investigation is

necessary to provide more comprehensive results regarding the effectiveness, ergonomics, and acceptability of sensory-friendly adaptation in video playback. The last limitation we faced was ethical constraints. The hypothetical scenarios used in interviews don't always accurately depict a particular scenario to its fullest extent. Unfortunately, these considerations limit the depth and validity of our results.

Since mimicking the adverse effects of intense light and sound may pose ethical concerns, a possible future direction could be to focus the participant pool on sensory-sensitive college students for a more informative and practical interview for researchers. Sensory-sensitive college students would have unique experiences with being triggered by sensory stimuli from electronic devices, so interviewing them to gain perspective for sensory-friendly design would not cause ethical concerns while obtaining valuable results. Another future direction would be to test whether our design aspects from the interview results are generalizable to a different population or to investigate the specific details regarding implementations of the sensory-friendly adaptation.

Enhancing technologies, diversifying participants, conducting real-world testing, and undertaking longitudinal studies while prioritizing ethical design is crucial to advancing sensory-friendly adaptability and digital well-being. For diversity, we must consider a broader range of participants from a wide range of demographics to ensure that the findings are generalizable. Longitudinal studies help us to understand the long-term impacts of sensory-friendly designs and their effect on digital well-being. Finally, ethical considerations should remain a top priority for researchers to bridge the gap between the needs of the user and the potential risks.

REFERENCES

- [1] Adjust the brightness and color temperature on your iPhone or iPad. Apple Support. (2022, November 8). <https://support.apple.com/en-us/HT202613>
- [2] Association for Computing Machinery(ACM). Designing Interactive Systems, 2022 June 13-17. Online (Asynchronous). 2022. Link: <https://dis.acm.org/2022/>
- [3] Emani Dotch, Jialuo Hu, Avery Mavrovounioti, Weijie Du, Jazette Johnson, Elizabeth Ankrah, Aehong Min, and Gillian R Hayes. 2023. Supporting Noise Sensitivity and Emotion Regulation with Children. In Proceedings of the 22nd Annual ACM Interaction Design and Children Conference (IDC '23). Association for Computing Machinery, New York, NY, USA, 522–526. <https://doi.org/10.1145/3585088.3593887>
- [4] Knoetze, Megan, et al. "Sound-level monitoring earphones with smartphone feedback as an intervention to promote healthy listening behaviors in young adults." *Ear and Hearing* 42.5 (2021): 1173-1182.
- [5] Meyer, J., McDowell, C., Lansing, J., Brower, C., Smith, L., Tully, M., & Herring, M. (2020). Changes in physical activity and sedentary behaviour due to the COVID-19 outbreak and associations with mental health in 3,052 US adults. 10.33774/coe-2020-h0b8g

- [6] Papamichael, Konstantinos, and B. Konstantinos. "Adaptive lighting for energy-efficient comfort and wellbeing." *SID Symposium Digest of Technical Papers*. 2017.
- [7] Rice, Mark, and Norman Alm. "Designing new interfaces for digital interactive television usable by older adults." *Computers in Entertainment (CIE)* 6.1 (2008): 1-20.
- [8] Roberts J, Yaya L, Manolis C. The invisible addiction: Cell-phone activities and addiction among male and female college students. *Journal of Behavioral Addictions*. 2014;3(4):254–265. doi: 10.1556/jba.3.2014.015.
- [9] Salvatore Saiu and Enrico Grosso. 2020. The SoundLight System: Fostering Adaptive Sound and Light Control in Clinical Environment. In *Proceedings of the 3rd International Conference on Applications of Intelligent Systems (APPIS 2020)*. for Computing Machinery, New York, NY, USA, Article 7, 1–6. <https://doi.org/10.1145/3378184.3378191>
- [10] Shinohara, K., & Wobbrock, J. O. (2016). Self-conscious or self-confident? A diary study conceptualizing the social accessibility of assistive technology. *ACM Transactions on Accessible Computing (TACCESS)*, 8(2), 1-31. <https://doi.org/10.1145/2827857>